



The Scientific Operations of Milani NavCam

Fabio Ferrari¹ · Iosto Fodde¹ · Felice Piccolo¹ · Carmine Giordano¹ · Antonio Rizza¹ · Alessia Cremasco¹ · Paolo Panicucci¹ · Lucia Francesca Civati¹ · Pietro Califano¹ · Mattia Pugliatti^{1,2} · Francesco Topputo¹ · Margherita Cardi³ · Marco Pavoni³ · Daniele Calvi³ · Andrea Zanotti³ · Filippo Corradino³ · Gabor Kovacs⁴ · Ernesto Palomba⁵ · Fabrizio Dirri⁵ · Chiara Gisellu⁵ · Marianna Angrisani⁵ · Paolo Tortora^{6,7} · Riccardo Lasagni Manghi^{6,7} · Edoardo Gramigna^{6,7} · Michael Küppers⁸ · Patrick Michel^{9,10}

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Abstract

Milani is one of the two 6U CubeSats on board the Hera spacecraft, a mission by the European Space Agency planned to rendezvous with the Didymos asteroid system in late 2026. The CubeSat will contribute to the scientific objectives of Hera, providing complementary observation geometries and close approaches to the asteroids. The large imaging dataset provided by Milani's Navigation Camera (NavCam) will contribute to measure global properties of the binary system and to characterize the asteroids' surfaces, within a spectral range between 400–700 nm. Both the spectral range and observation geometry are complementary to other imaging instruments on board Milani and Hera. In addition, the NavCam will be instrumental to the AutoNav experiment, the main technological objective of Milani.

Keywords Milani · Hera · Didymos · DART · Planetary defense · Asteroid · CubeSat

1 Introduction

The European Space Agency's Hera mission will be the first to rendez-vous with a binary asteroid system. Hera was launched on October 7, 2024 and will arrive at (65803) Didymos in late 2026. Upon arrival, Hera (Martino et al. 2026) will release two 6U CubeSats, Milani (Cardi et al. 2026) and Juventas (Karatekin et al. 2026), which will contribute to the scientific objectives of Hera (Michel et al. 2025), and to expand its scientific output, by providing independent measures of the properties of asteroid Didymos and Dimorphos. Milani CubeSat was developed by the company Tyvak International in Italy, leading a consortium of entities from Italy, Czech Republic, Finland.

Hera and its CubeSats will be the first to visit Didymos after NASA's DART (Daly et al. 2023) spacecraft impacted Dimorphos on September 26, 2022 and the first multi-satellite system devoted to deep space exploration of asteroids. This will provide the unique opportunity to measure in-situ the effects of a hypervelocity impact on a rubble-pile asteroid, and to study the properties of the binary system post-impact. To address these important scientific questions, Milani is designed to follow a challenging mission profile, which involves

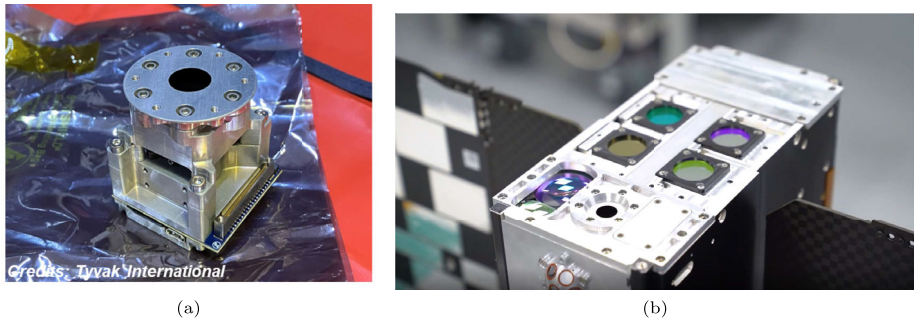


Fig. 1 (a) NavCam (with baffle) on the test bench and (b) integrated into Milani's optical deck at Tyvak's integration facility

the execution of several maneuvers. The Navigation Camera (NavCam) of Milani is primarily designed to serve as main navigation instrument, to provide optical images of Didymos and Dimorphos, then used for trajectory and attitude reconstruction of the CubeSat. At the same time, these images hold a great scientific value, as they provide information about the global properties, surface, and dynamics of both Didymos and Dimorphos. The NavCam is a RGB camera, covering a spectral range between approx. 400 – 700 nm. This is complementary to spectral ranges provided by other imaging instruments aboard Milani (Näsilä et al. 2025) and Hera (Vincent et al. 2026; Popescu et al. 2025). Also, it will provide different observation geometries compared to what is available on Hera. RGB images can be directly compared with images taken by the LUKE's RGB camera of LICIACube CubeSat (Dotto et al. 2024; Poggiali et al. 2022), right after DART's impact on Dimorphos. In addition, the NavCam will be used to demonstrate the technological objective of Milani, regarding the autonomy of CubeSat in deep space. A more detailed discussion on the expected scientific output of the NavCam, and the autonomous navigation experiment are reported in Sect. 4.

2 Overview of the Milani NavCam

The NavCam is based on a Tyvak Star Tracker technology, customized for the Milani mission. The NavCam provides images of Didymos and Dimorphos and has a twofold purpose. First, to provide guidance to the autonomous attitude pointing mode of Milani's Attitude Determination and Control System (ADCS, Pugliatti et al. 2023c, 2024). Second, to contribute to Hera and Milani mission and scientific objectives, providing global maps of the surface of both asteroids in the visible spectrum. The NavCam is shown in Fig. 1, and is made of the optical head, the detector, and a baffle. The NavCam's optical head was designed and coated specifically for Milani by Optec SpA. It is composed of three lenses as shown in Figs. 2a and 2b, ensuring a focus at infinity with a maximum of 1% distortion. The optical head is designed to work between 30 and 0.2 km, which is largely compliant with the observation distance occurring during the mission (see Sect. 3). The optical design ensures an average transmissivity above 90% in the visible spectrum (i.e., 400 – 700 nm) and a focal length of 12.96 mm. The maximum wavelength (i.e., 700 nm) is imposed by applying an external coating to the first lens. This shields the detector from near-infrared light, given the poor but non-null response of the detector in this bandwidth and the overlap of the spectral response of the three channels.

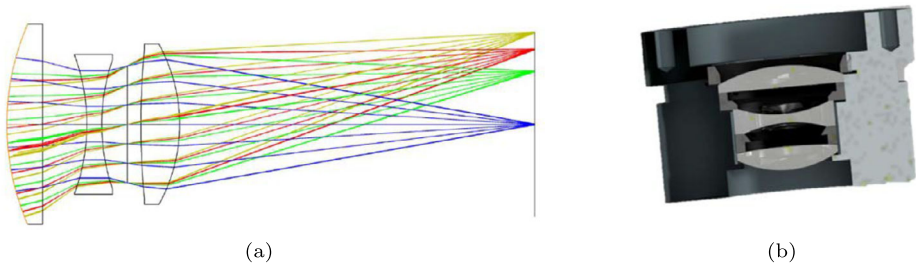
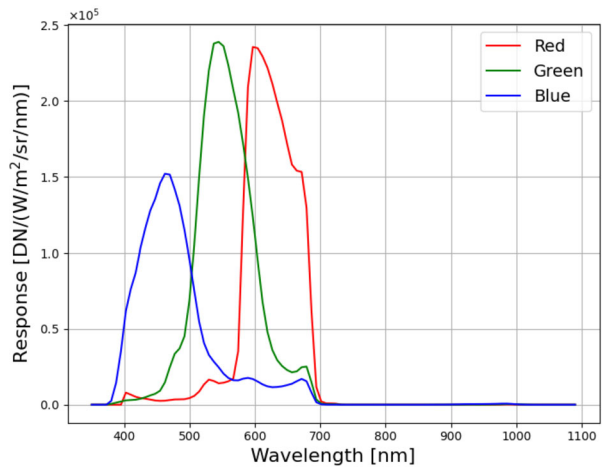


Fig. 2 Schematics of NavCam's (a) optical and (b) mechanical design

Fig. 3 The spectral response of the different RGB channels of the NavCam for a reference exposure time of 0.005 s



The detector is a 1/3-inch CMOS rolling-shutter digital image sensor manufactured by Onsemi, which was chosen due to its versatility to serve both Milani's engineering and scientific objectives, and has heritage from other space systems from Tyvak, the payload provider. It is equipped with a RGB Bayer filter to provide color information for scientific investigation. In principle, the detector is capable of measuring wavelengths up to 850 nm. However, the S/N ratio is low for wavelengths above 700 nm, and therefore, to avoid additional noise, an optical coating was added to the first lens to filter out wavelengths above 700 nm. The final response curves for the individual colors, after the radiometric calibration (discussed later in this section), are shown in Fig. 3. It is worth noting that color data are only used for scientific purposes and are not used for navigation, for which the algorithm is agnostic to detailed spectral variations. Navigation is centroid-based, and the image processing algorithm makes use of all illuminated pixels above a certain threshold to estimate the centroid of the asteroid. The detector consists of a matrix of 2048×1536 active square pixels of $2.2 \times 2.2 \mu\text{m}$. The low pixel pitch ensures an instantaneous Field of View (iFoV) of 35 arcsec, which translates into a Ground Sampling Distance of 169.75 cm at 10 km and 8.48 cm at 500 m. The NavCam has a Field of View (FoV) of 19.72×14.86 degrees. A summary of the NavCam properties is reported in Table 1.

The NavCam was calibrated at the Budapest University of Technology and Economics (BME), as the other instruments on board Hera. The calibration process included: bias and dark images, identification of bad pixels, flat-field images for each color, a linearity test, a radiometric response test, and a distortion test. The detector has several internal image

Table 1 Properties of the Milani NavCam

Parameter	Value
Field of View [deg]	19.72 × 14.86
Field of View [m]	174 @ 500 m distance 695 @ 2 km distance 3476 @ 10 km distance
Spectral range [nm]	400–700
Image size [pixels]	2048 × 1536
Pixel size [μm]	2.2 × 2.2
Focal length [mm]	12.96
F-number [-]	2.8
Average lens transmissivity [-]	90%
Working distance [km]	30–0.2
Resolution [cm/px]	8.48 @ 500 m distance 33.95 @ 2 km distance 169.75 @ 10 km distance

processing capabilities. By default, the sensor performs black level correction using a set of dark pixels to calculate the dark current, and a pixel defect correction is also applied on-chip. Besides this, a high dynamic range mode can be enabled which uses gamma correction and a binning technique on-chip. However, these are not used in the calibration and performance test discussed here. Both the dark and bias images show a uniform noise pattern with a very low standard deviation (~ 0.6 digital numbers (DN)). The dark images did not show an increase in noise as a function of exposure time, indicating that a dedicated dark calibration step might not be needed. This is likely a result of the black level correction happening on-chip. Both the dark and bias frames were obtained at a single temperature. Even though this temperature is close to the operating range, in-flight dark and bias calibration images are still needed to confirm if there is a temperature dependence or not. On-ground, no bad pixels were identified (possibly due to the on-detector post-processing capabilities), which will be updated in-flight when bad pixels are identified in sequential images. The flat field showed largely uniform responses, except for some strong vignetting (50 percent drop in intensity towards the edges). Both the vignetting and the low amount of dust contamination on the optics were found to be completely removable using a flat field correction. In particular, the flat field correction provides better than 1% uniformity. The absolute radiometric uncertainty is better than 2.5%. However, this is strongly dependent on the illumination, and the spectral characteristics of the pixel Bayer filters, as the flat correction has a broad-band effect. The distortion correction will be best estimated in flight, as ground tests imply atmospheric environment and a finite object distance of the test chart, resulting in slight blurring. For a more accurate test, in flight starfield images should be acquired with 5–10 sec exposure times and match the star positions with the Hipparchos or Tycho catalogs. The current result uncertainty is about 1.5 pixels. The improved starfield calculations can result 0.1-pixel uncertainty. The linearity test showed a different response for each color channel as a function of exposure time. Mainly during very low exposures, below 1.0 ms, the linearity error is large. Besides that, the R and G channels show in general a good linearity trend, but the blue channel has relatively large linearity errors for exposures higher than 0.008 seconds (likely due to the on-board gamma correction). The radiometric calibration resulted in a conversion between DN and radiance units as shown in Fig. 3. Finally, the distortion tests showed low but recognizable distortion effects, especially near the edges the distortion could lead to errors in the order of 50 pixels. With a radial distortion correction this can be largely corrected

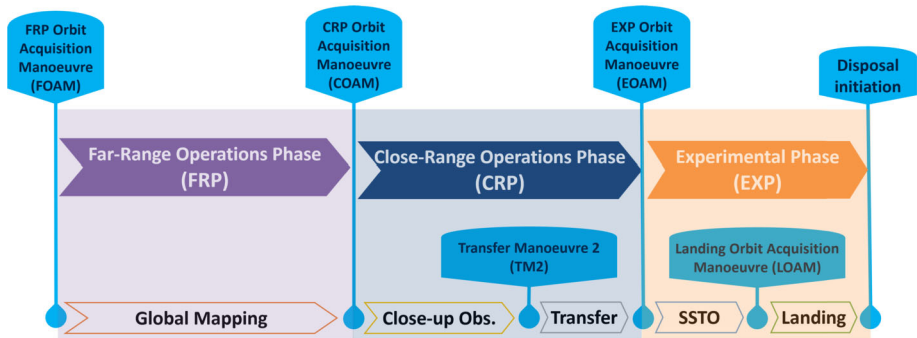


Fig. 4 Operational scientific phases of Milani

for. Figure 3 provides the experimental value for the full camera transfer function, including optics and sensor quantum efficiency. From the datasheet, the peak sensor QE for the different color channels are Red - 55%, Green - 62%, and Blue - 48%. From the on-ground calibration it was found that the dark images show around a mean value of 166.9 DN with a 0.6 DN noise (12 bit encoding), with no significant increase with respect to exposure time or temperature. We do not expect the need for stacking of images, given that the asteroid is resolved and the exposure time can be changed to adapt to the asteroid albedo variation. Anyhow, stacking (and coadding) may be envisioned during operations for dedicated need (e.g., particle detection).

3 Operations

3.1 Mission Phases

The design of Milani's trajectory was driven by the scientific objectives of the mission, while also being affected by technical and operational constraints. The sequence and timeline of all mission phases are reported in Cardi et al. (2026), while Fig. 4 reports the breakdown of the scientific phases only, where Milani will be operational. The nominal mission profile includes two main phases, which are designed to achieve all mission and scientific objectives of Milani, followed by an opportunity experimental phase:

- **Far-Range Operations Phase (FRP):** this phase lasts for approximately 3 weeks and aims to acquire a global mapping of the asteroids;
- **Close-Range Operations Phase (CRP):** lasting roughly 4 weeks, the CRP aims to perform close-up observations of Dimorphos and to observe the DART impact site;
- **Experimental Phase (EXP):** Milani will be inserted into an Self-Stabilized Terminator Orbit (SSTO) and will gradually descend towards the Didymos system, eventually attempting a landing on either Dimorphos (currently baselined) or Didymos.

The nominal orbits of Milani during the FRP, CRP, and EXP are reported respectively in Figs. 5, 6, and 7. All trajectories are shown in a frame centered at the Didymos Barycenter (except for Fig. 7b, which is centered at the barycenter of Dimorphos), with the x-axis directed towards the projection of the Sun in the Didymos equatorial plane, the z-axis aligned with the spin axis of Didymos, pointing towards its south pole, opposite to the orbital angular momentum of Dimorphos, and the y-axis completing the right-hand coordinate system.

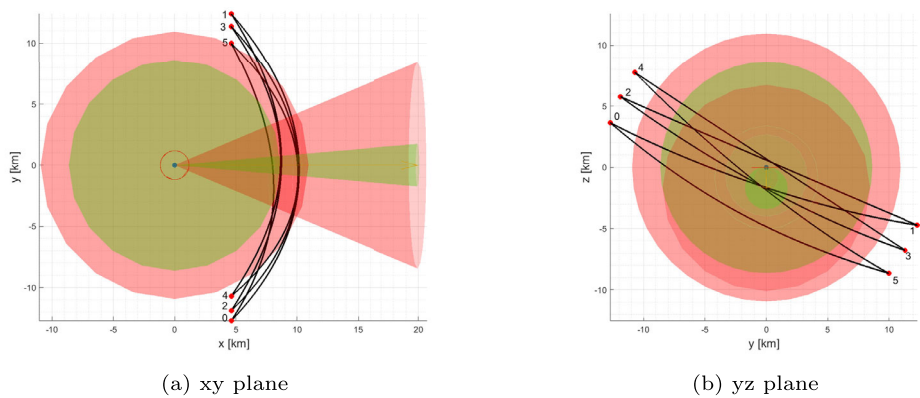


Fig. 5 FRP trajectory (solid black line), consisting of alternating 4-day and 3-day arcs, repeated three times. Science admissible regions for global mapping are highlighted in terms of range (8.64–10.94 km, green and red spheres) and phase angle (5–25 deg, green and red cones)

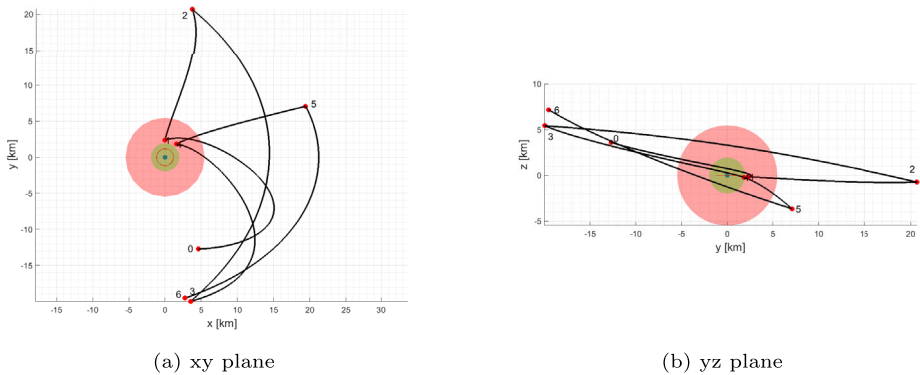


Fig. 6 CRP science orbit (solid black line). 7-point loop with a 7-4-7-7-3-7 days maneuvering pattern. Science admissible regions for global mapping are highlighted in terms of range (1.96–5.48 km, green and red spheres). Phase angle cones are not shown here

Suitable regions for scientific observations are highlighted in Figs. 5, 6 in terms of angular boundaries (min-max Sun phase angle, green-red cones) and ranges (min-max ground resolution, green-red spheres). More details about the trajectory design criteria are reported in Ferrari et al. (2021b,a), Bottiglieri et al. (2023). More insights about quantitative requirements regarding phase angles and resolution are reported in figure captions and detailed in Cardì et al. (2026).

3.2 Data Acquisition Profile

We report here the observation geometry of the NavCam over time, compared with that of Hera and Juventas (ESA 2024), and the other imaging instruments: Hera's AFC, Milani's ASPECT, Juventas' NavCam. Figures 8, 9, 10, and 11 show, respectively, the range with respect to Didymos and Dimorphos, the phase angles, the apparent sizes of the asteroids, and the Ground Sampling Distance (GSD) of Didymos and Dimorphos. All figures are plotted

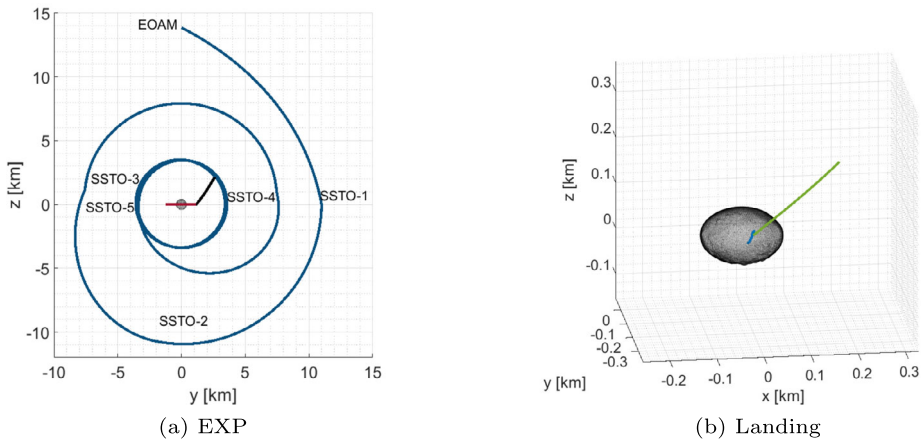


Fig. 7 Self-Stabilized Terminator Orbit (SSTO), gradually approaching the Didymos system, and eventually landing on the surface of Dimorphos

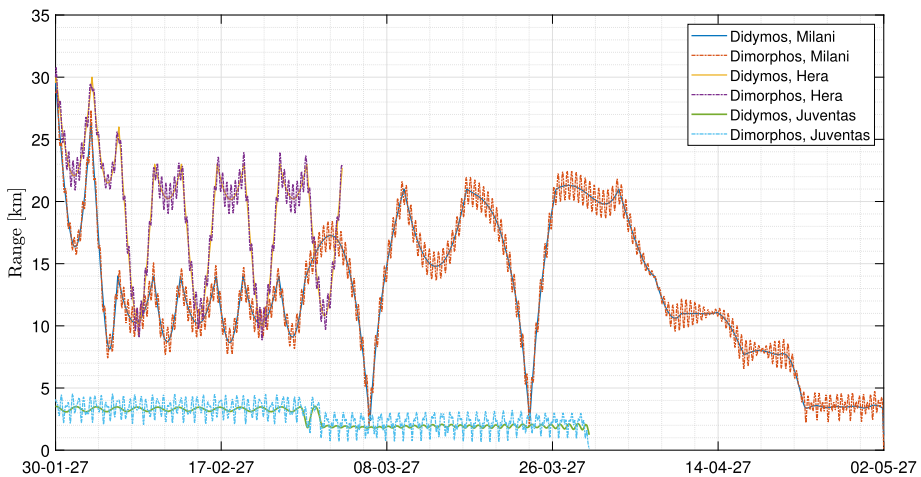


Fig. 8 Range of Milani, Hera, and Juventas with respect to Didymos and Dimorphos throughout Milani's entire mission profile

against Milani's mission timeline, which is currently planned to start on 30 January 2027, when the Cubesat will be released from the Hera Spacecraft. All trajectory information is retrieved from the latest and most up-to-date Hera Mission Kernels (ESA 2024). Please note that the trajectories of Milani and Juventas reported in the current version of the kernels refer to a later release date. In the aforementioned plots, their trajectories have been shifted in time to match the baseline deployment date for consistency. Additionally, Hera's trajectory is currently available only up to the end of its Detailed Characterisation Phase (DCP).

Milani NavCam provides unique viewing geometries compared to Hera and Juventas, offering complementary scientific observations and improved measurement accuracy. Figure 12 shows the cumulative GSD of Didymos and Dimorphos, illustrating the amount of time spent below specific GSD thresholds. For example, the Milani NavCam can operate for

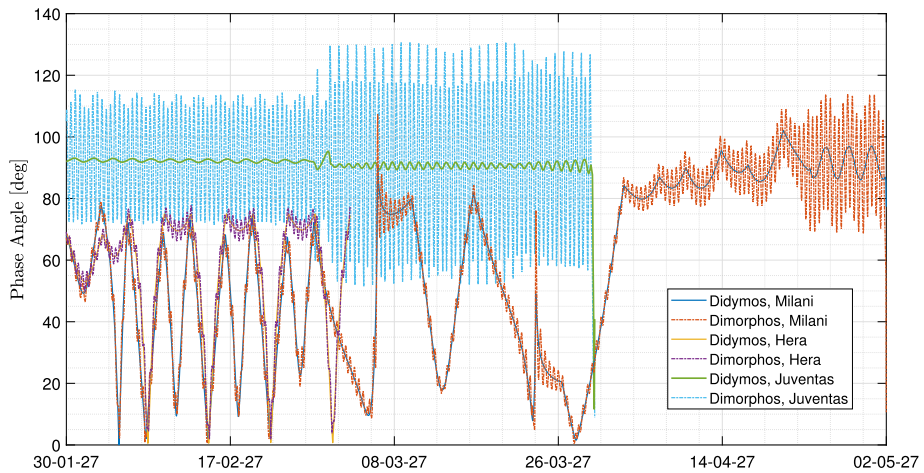


Fig. 9 Phase angles of Didymos and Dimorphos with respect to Milani, Hera, and Juventas throughout Milani's entire mission profile

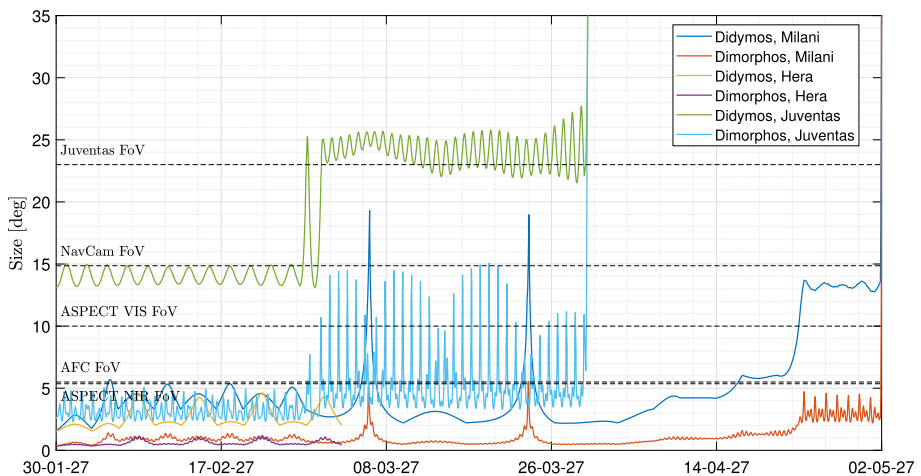


Fig. 10 Apparent size of Didymos and Dimorphos with respect to instruments on Milani (NavCam, ASPECT), Hera (AFC), and Juventas (NavCam) throughout Milani's entire mission profile

37 days with a GSD of less than 2 m/pixel for Dimorphos, while Hera can achieve this for 20 days and Juventas for 61 days. Although in a shorter time frame, Fig. 9 shows that Milani can acquire images at a variety of phase angles, which is useful for surface characterization, while Juventas' phase angle is more limited due to its SSTO trajectory. Note that Hera's trajectory is considered here only until the end of the DCP. During the subsequent phases, the Hera spacecraft will likely approach the asteroids, and therefore its cumulative GSD is expected to improve at finer resolutions.

NavCam images represent a significant portion of the data budget. Indeed, around 200 MB of asteroid images will be downlinked after applying JPEG2000 lossless compression onboard for orbit determination, flight dynamics, and scientific purposes. As a result,

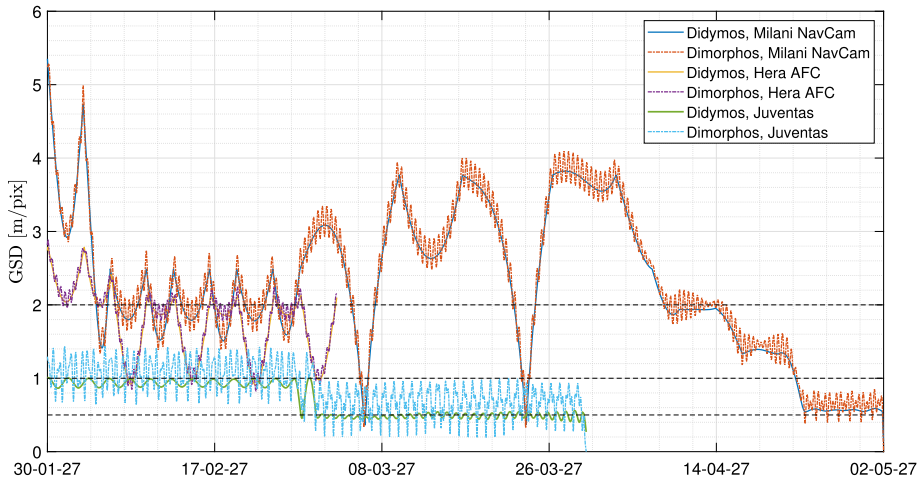
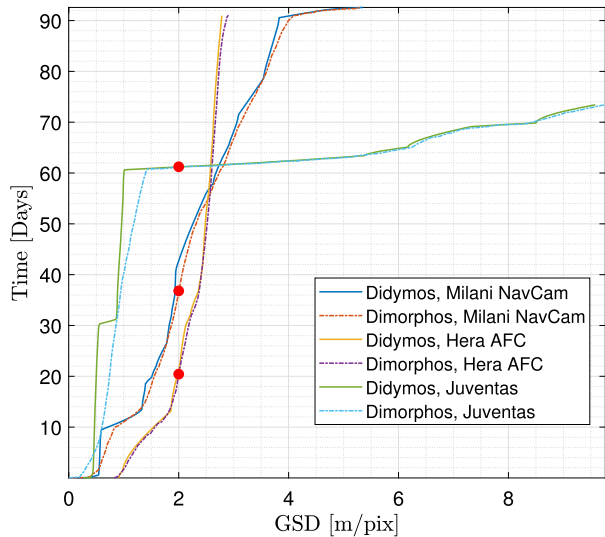


Fig. 11 GSD of Didymos and Dimorphos with respect to instruments on Milani (NavCam), Hera (AFC), and Juventas (NavCam) throughout Milani's entire mission profile. The dashed lines refer to Milani NavCam's scientific requirements on surface mapping and imaging (images to be acquired at resolutions better than 2, 1, and 0.5 m/px)

Fig. 12 Cumulative GSD of Didymos and Dimorphos for instruments on Milani (NavCam), Hera (AFC), and Juventas (NavCam). The red dots represent the cumulative GSD of Dimorphos below 2 m/px



approximately 540 images will be downloaded to ground during the entire lifetime of Milani mission. On average, this results in one NavCam image every 4 hours. Figures 13, 14, and 15 show examples of NavCam acquisition for Didymos and Dimorphos at a resolution of 2 m/px, 1 m/px and 0.5 m/px respectively. Synthetic images are generated using pre-impact shape models of Didymos and Dimorphos (ESA 2020) and rendered using CORTO (Pugliatti et al. 2023a).

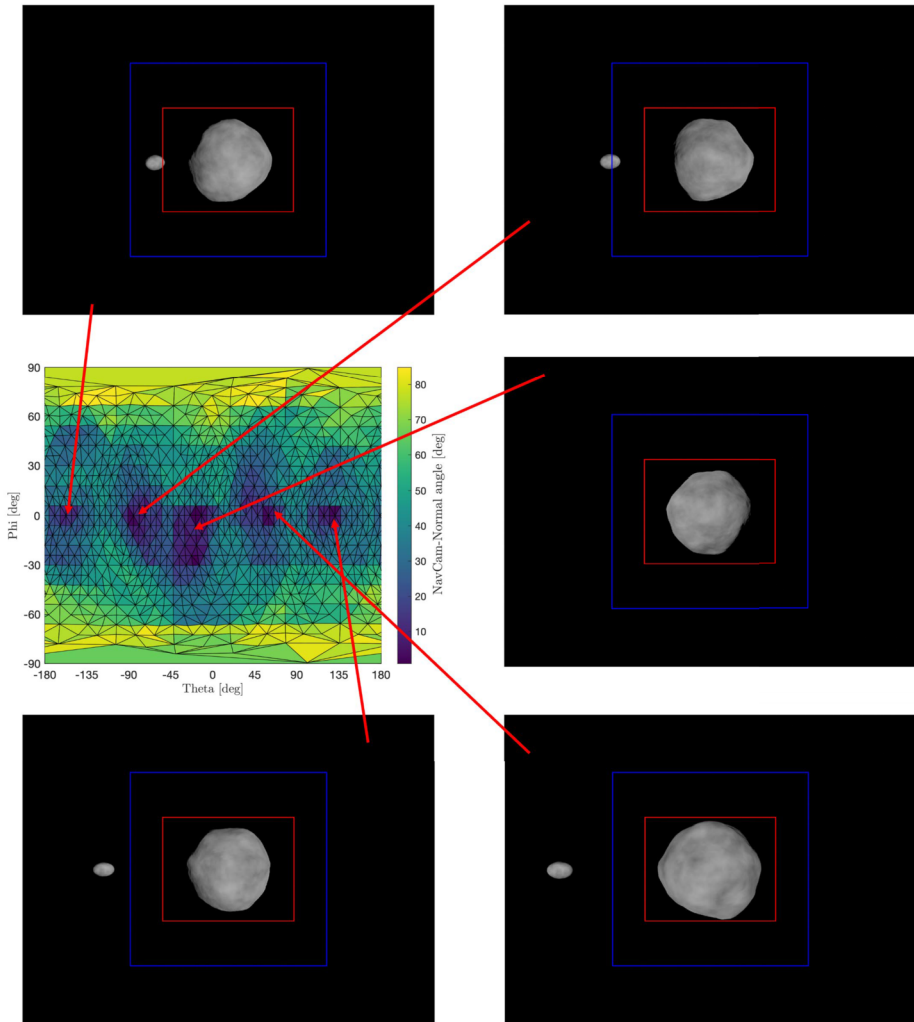


Fig. 13 Example of observation sequence for Didymos at 2 m/px (FRP) with 5 acquisitions. The images are sized with the FoV and resolution of the Milani NavCam. For reference, the blue and red rectangles represent respectively the VIS and NIR FoV of ASPECT. The middle left graph shows the map of the best (i.e., minimum) incident angle between the Milani NavCam boresight and the Didymos surface normal among the five-image observation sequence. Theta and Phi represent the longitude and latitude of Didymos, respectively

4 Expected Scientific Output

The NavCam will contribute to the main scientific objectives of Milani. A general overview of the Science Traceability Matrix (STM) associated with the NavCam is shown in Table 2. The left column of the STM provides several of the Milani scientific and technological goals. The middle column represents the NavCam objectives that will contribute to the general Milani goals, while the final column lists the requirements for the NavCam regarding the specific measurements it will need to obtain to reach the related objectives. A more detailed

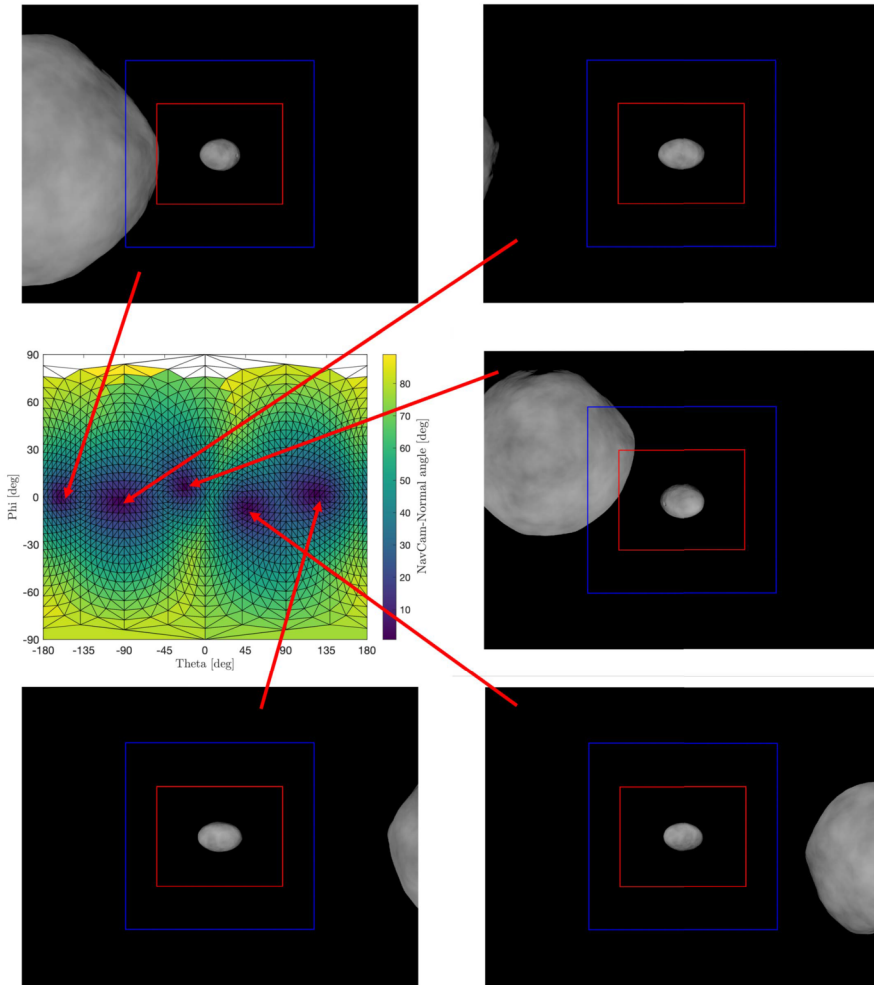


Fig. 14 Example of observation sequence for Dimorphos at 1 m/px (CRP) with 5 acquisitions. The images are sized with the FoV and resolution of the Milani NavCam. For reference, the blue and red rectangles represent respectively the VIS and NIR FoV of ASPECT. The middle left graph shows the map of the best (i.e., minimum) incident angle between the Milani NavCam boresight and the Dimorphos surface normal among the five-image observation sequence. Theta and Phi represent the longitude and latitude of Dimorphos, respectively

version of the STM for the Milani mission is reported in Cardi et al. (2026). In the next section, individual objectives and requirements will be discussed in more detail.

4.1 Global Properties

The global properties of Didymos and Dimorphos are macroscopic parameters like mass, shape, density, rotational state, and porosity, that physically define the system and its dynamical evolution. The shape of both Didymos and Dimorphos will allow for various scientific questions to be answered. Before the DART arrival, the shape of Didymos was thought

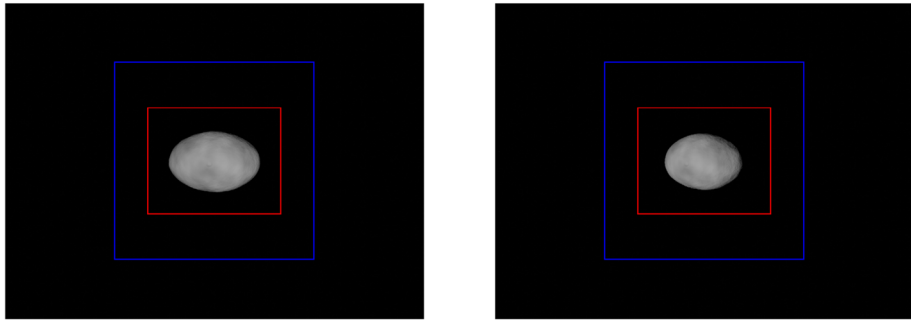


Fig. 15 Left: example of DART impact site acquisition at phase angle 7.29 deg. Right: example of DART impact site acquisition at phase angle 33.22 deg. The images are sized with the FoV and resolution of the Milani NavCam. For reference, the blue and red rectangles represent respectively the VIS and NIR FoV of ASPECT

Table 2 The science and technology traceability matrix for the Milani NavCam

Milani Goals	NavCam Objectives	NavCam Requirements
The mission shall map the global composition of the Didymos asteroids.	To support the determination of the global properties of the Didymos asteroids.	The surface of the Didymos bodies shall be mapped to achieve at least a 90% global coverage, except areas which might be permanently shadowed during the mission. The shape of the Didymos bodies shall be determined.
The mission shall characterize the surface of the Didymos asteroids.	To observe the composition of the Didymos bodies surface.	The NavCam shall improve the rotational model of both asteroids. The phase curves of selected areas of the surface shall be measured.
The mission shall identify local shock effects on Dimorphos caused by DART.	To observe the morphological and spectroscopic effects of the DART impact.	To compare mature and freshly exposed material on a global scale. The diameter of the crater (if present) shall be measured. To compare shocked and unshocked material at global scale. The difference in shape, or any impact-produced surface characteristic, with respect to the pre-DART impact conditions shall be measured.
The mission shall support Didymos gravity field determination.	The mass measurement of both asteroids, and the accuracy of higher order gravity field parameters (J2 and higher) shall be improved.	The NavCam shall provide additional measurements to improve the gravity field determination of both bodies.
The mission shall demonstrate the utilisation of CubeSat technologies in deep space.	Validate optical-based autonomous navigation algorithms.	Autonomous optical navigation algorithms shall be implemented and used with images produced by the NavCam.

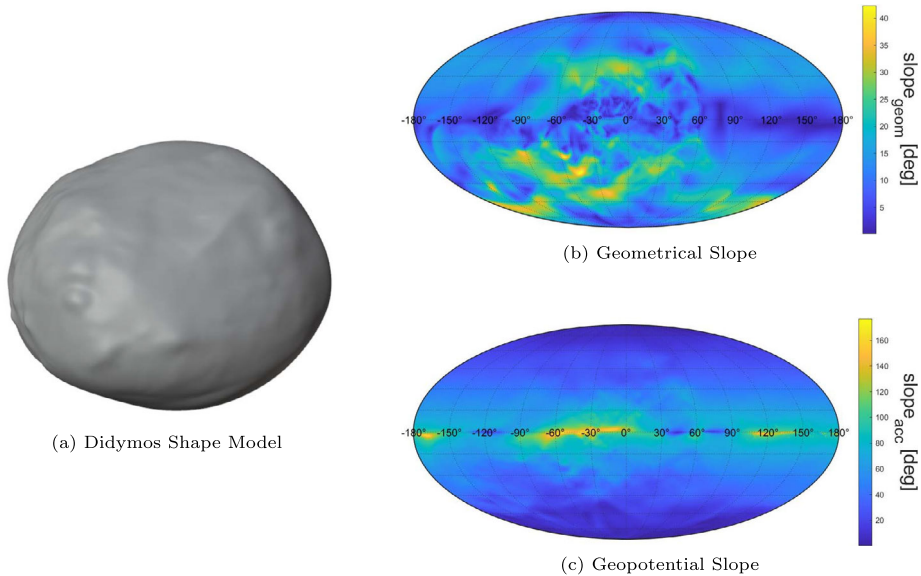


Fig. 16 The shape and slopes of Didymos, taken from the latest shape model of Didymos from DART data (Barnouin et al. 2024)

to be very similar to asteroids Bennu and Ryugu, i.e. a top-shaped asteroid. This raised the question of what processes guide the generation of these shapes, as their spectral types, and thus their composition, are different (B / C type compared to S type, Michel et al. 2022). The arrival of DART and the images taken by the DRACO camera revealed a shape that was significantly different than previously estimated (Barnouin et al. 2024), with a much more pronounced equatorial bulge and a shorter length along the polar axis. The updated shape of Didymos can be seen in Fig. 16a. The shape of Didymos, together with its measured rotation, has significant implications on the formation of binary asteroids as it can hint towards certain formation scenarios (Wimarsson et al. 2024). Obtaining high-resolution maps of the full body (current shape model is biased in resolution towards the side that was observed by DRACO) will allow us to better infer how the shape of Didymos is linked to the formation of binary asteroids. Combining the shape with mass and interior measurements made by JuRa (Herique et al. 2026) will also allow for density estimates of both Didymos and Dimorphos, better constraining their dynamical history and allowing for estimating the similarities between the two bodies. A heterogeneous density distribution can also be measured by comparing the calculated gravity properties of a constant density model with the measured gravity field (Scheeres et al. 2020). Another type of process that significantly impacts the evolution of (binary) asteroids is the (binary) Yarkovskii-O'Keefe-Radzievskii-Paddack ((B)YORP) effect (Čuk et al. 2024), which alters the rotation and orbital state of the system over long periods of time due to thermal radiation. This effect is highly dependent on the shape of the asteroid, and thus a highly detailed shape model will improve our understanding of the precise influence of this effect on the evolution of this system. Finally, the shape of Dimorphos will likely be significantly altered due to the impact of DART (Raducan et al. 2024). A detailed shape model of Dimorphos will allow for the determination of how effective the kinetic deflection exactly was, as it will reveal information on various properties of Dimorphos, e.g. internal strength, stiffness, internal structure, etc.

One of the main scientific objectives of Milani is to provide the global map and shape of both Didymos and Dimorphos. To do so, the NavCam will acquire images at several different illumination conditions and phase angles. This will provide both an independent estimate of global properties of the asteroid system, to be compared and validated against Hera-based instrumental estimates, and a way to refine and complement such measures, as Hera and Milani will be observing the asteroids with different geometries and illumination conditions. The NavCam will therefore contribute to Hera's main shape model derivation, which will be led by Hera's Asteroid Framing Camera (AFC) team (Vincent et al. 2026). Together with the images from the other instruments, this will lead to a detailed shape models of both Didymos and Dimorphos. Shape models will be produced using stereophotoclinometry (SPC, Adam et al. 2023). This technique combines observations made from the limb of the body when the body is not fully in the field of view of the camera, with images from individual maplets of the surface containing landmarks. These maplets, and their 3D position obtained through triangulation, are combined with the limb-based data to generate a high resolution digital terrain model (DTM). A more in-depth explanation of how this process works is given in Palmer et al. (2022). Using both the SPC method and the tracked landmarks within the maplets, the rotational state of both bodies can also be estimated (Panicucci et al. 2023). The different requirements on observation geometry for the NavCam are based on the optimal parameters for shape model generation through SPC.

4.2 Surface Characterization

The scientific observations of the surface will consist of two parts: its geomorphological properties and its spectral properties. The NavCam will provide a global map of the surface, and this will support a full geomorphological characterization of both Didymos and Dimorphos. In particular, one of the first investigation that can be done is to look at the surface slopes of both bodies. A map of the surface slopes derived from the current shape model of Didymos can be found in Fig. 16b. The slope map can be combined with the estimates of the rotational state and gravity field to obtain a geopotential slope map, which measures the angle between the (negative) surface normal and the local acceleration vector, see Fig. 16c. By obtaining observations of regolith movement and slope destabilization, and correlating them with the slope maps, surface characteristics like regolith properties, gravity anomalies, and density distribution discrepancies can be estimated. Additional investigations include measurement of grain and boulder size frequency distribution on the surface, as well as estimating properties and shapes of surface material, or searching for specific surface features, such as thermal cracks, ridges or craters. For spectral investigations, several different properties can be measured. First, lab tests have shown that shocking certain materials can have a large effect on their spectral properties (Kohout et al. 2020). Obtaining information on the spatial distribution of the shocked material will allow for a better understanding of what happened after the DART impact. Furthermore, shocked material can also indicate natural impacts, providing information on the age of the system. Similar changes in spectra can be observed for weathered and unweathered material (Kohout et al. 2020). Besides also helping in determining material that was recently exposed due to the DART impact it also allows for unprecedented investigations into the process of space weathering and how this affects the transport of different materials across the Solar-System (Clark et al. 2023). Finally, measuring the compositional properties of the asteroid will allow for inferring a meteorite analogue of the system, which in turn will allow for better bulk density and porosity estimate.

The RGB sensor allows for measuring the incoming light through three different filters, with their spectral responses shown in Fig. 3. To test how well the NavCam can detect

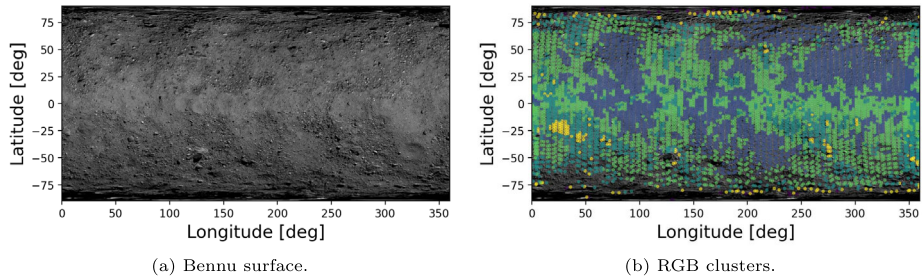


Fig. 17 Example results from RGB clustering across the surface of Benu, where spectra measured by the OVIRS instrument on board OSIRIS-REx were taken and converted to RGB measurements. The different colors represent the different clusters found by the k-means algorithm

surface heterogeneity and composition, the sensor response can be simulated for known materials. Given a spectrum corresponding to a certain material, the spectrum needs to be convoluted with the RGB sensor response curves to obtain the different R, G, and B outputs, as follows:

$$I_c = \frac{t_{\text{exp}}}{t_{\text{ref}}} \int_{\lambda_{lb,c}}^{\lambda_{ub,c}} S_m(\lambda) R_c(\lambda) d\lambda, \quad (1)$$

where I_c is the sensor reading for channel c , $\lambda_{lb,c}$ and $\lambda_{ub,c}$ are the channel's wavelength lower and upper bound, $S_m(\lambda)$ is the material spectrum in radiance units ($\text{W}/\text{sr}/\text{m}^2$), and $R_c(\lambda)$ is the channel's response function in units $\text{DN}/(\text{W}/\text{sr}/\text{m}^2)/\text{nm}$ (see Fig. 3). The response function is an experimentally obtained value, including several aspects of the camera response like detector quantum efficiency and optical transfer function, obtained at a reference exposure time $t_{\text{ref}} = 0.005$ s. In the next analyses it is assumed that the exposure time $t_{\text{ref}} = t_{\text{exp}}$.

As a test case, the surface of Benu is used to determine how well the NavCam RGB measurements perform when detecting surface heterogeneity. The surface of Benu was mapped spectroscopically by the OVIRS instrument on board the OSIRIS-REx spacecraft (Reuter et al. 2018). Calibrated data was taken from the detailed survey phase and filtered for observations with fill factor equal to 1 and incidence and emission angles lower than 70 degrees. Using Eq. (1), the spectra at the different wavelengths were then converted to theoretical RGB values that the Milani NavCam would observe. These measurements can then be clustered using statistical clustering algorithms, as was done for various previous missions (Perna et al. 2017; Barucci et al. 2020). Figure 17 shows the results when applying a k-means clustering algorithm (MacQueen 1967) to the simulated NavCam RGB measurements for Benu. Several different distinct groups were found within the data, which when compared to previous results (Barucci et al. 2020), show some overlap with their spectroscopic mapping. This shows the potential for NavCam measurement to detect the different surface materials on Didymos and Dimorphos. Other clustering techniques, such as the G-mode analysis (Poggiali et al. 2022), where no a priori number of clusters needs to be set, will be explored in future work.

The different clusters have certain associated RGB value distributions. To estimate how these different RGB values relate to the composition of surface elements, the response of different materials first has to be simulated using lab data. For this, a database of spectra was used called RELAB (Pieters and Hiroi 2004). This database contains a large amount of different minerals and meteorites, with also shock and space weathering experiment results

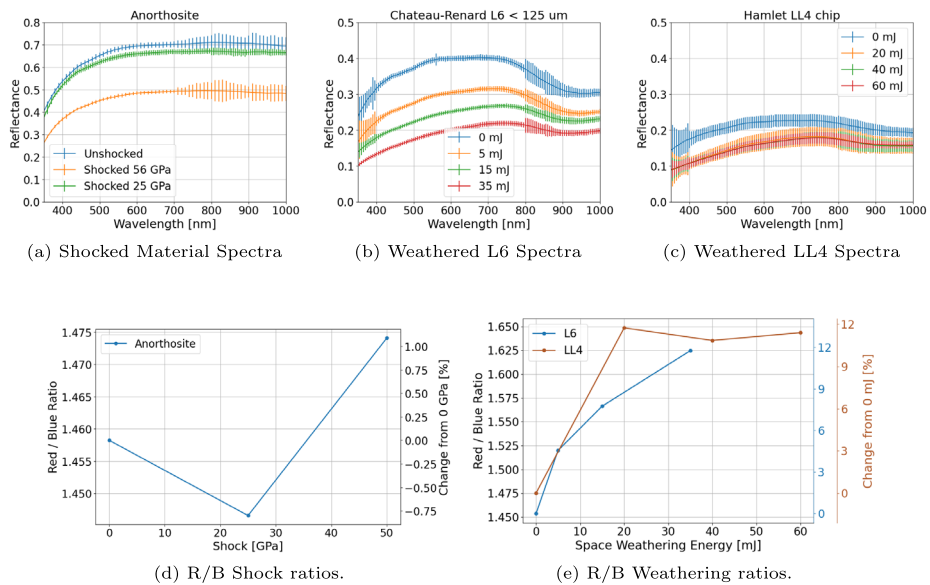


Fig. 18 Different spectra for shocked Anorthosite and weathered L6 Chateau-Renard ordinary chondrite meteorite (a) - (c), and how the red over blue ratio changes as a function of unshocked/unweathered and shocked/weathered material (d) - (e)

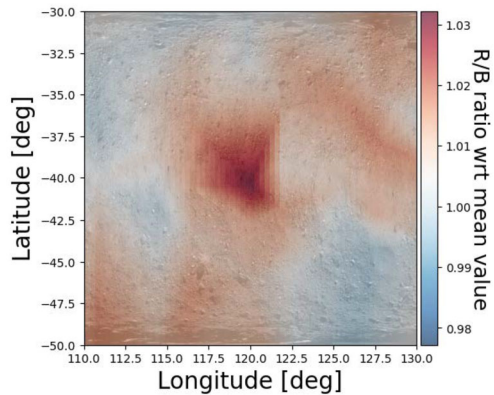
present in the database. A set of these spectra with their respective uncertainty bounds can be seen in Figs. 18a - 18c. Again, using Eq. (1), the simulated NavCam RGB measurements related to these materials can be determined.

Figure 18d shows how the simulated red to blue ratio changes for the Anorthosite sample for different shock values. The ratio only changes by roughly a percent, which is likely hard to observe in actual measurements. This is expected from the spectrum shown in Fig. 18a, which shows that shocking the material changes its reflectance equally for all wavelengths, thus not changing its color. More experiments with different samples should be performed to determine the observability of shocked material. For weathering, the results are shown in Fig. 18e. In this case, the changes when the material is irradiated are much clearer, as the material becomes redder and the red over blue ratio increases by roughly 12 percent. Therefore, the camera is likely capable of measuring surface weathering processes by observing the distribution of the RGB ratios across the surface. An example of the science that can be performed by these measurements is given in Fig. 19, where the simulated red over blue ratio is shown for a patch of surface of Bennu, where again the OVIRS spectra are taken to generate the simulated measurements. It is clear that by measuring these ratios and comparing different locations on the surface, regions of possible space weathering can be identified. Combining these measurements from the NavCam with other spectral instruments on board Milani and Hera, like ASPECT and Hyperscout, can thus improve the overall surface characterization of Didymos and Dimorphos.

4.3 Support to Gravity Field Determination

As mentioned in Sects. 4.1 and 4.2, the gravity field is critical to various geophysical investigations. Asteroid gravity gives insights into the internal structure, surface morphology,

Fig. 19 Example observation of surface patch of Bennu using OVIRS spectra, where the simulated red over blue ratio with respect to the mean surface value is shown



and homogeneity of the internal mass distribution. Moreover, the gravity field provides information about the dynamic environment around the asteroid and on its surface, providing clues about the asteroid's geological evolution and resurfacing. Determining the gravity field is important not only to understand the asteroids' history but also to foresee the trajectory evolution of the spacecraft and dust orbiting the system. The University of Bologna leads the gravity field determination for the Hera mission, and is responsible for the Hera Radio Science Experiment (RSE, Gramigna et al. 2024; Tortora et al. 2025). The RSE will exploit measurements from several instruments, namely radiometric tracking data from the Earth, inter-satellite link data between Hera and the CubeSats, and optical images from Hera's Asteroid Framing Cameras (AFC). These measurements will be processed within a batch least-square filter as part of an orbit determination (OD) process, resulting in the joint estimation of the spacecraft motion and of key physical parameters of the system, including the asteroids' mass, rotational states, moments of inertia, and higher-order gravity harmonics.

A key factor in improving the accuracy of the gravity field estimate is the time spent by the spacecraft in close proximity to the asteroids. Indeed, the influence of higher-order harmonics is more noticeable close to the surface. Furthermore, a reliable global estimate requires that the spacecraft dynamics is constrained from various locations in the asteroid's body frame, maximizing ground track coverage. In this regard, Milani's NavCam images could provide useful information by complementing the AFC's images from different orbital geometries. Furthermore, tracking data acquired in the EXP phase during descent and landing can provide valuable constraints to model the local gravity field near Dimorphos.

The NavCam information content for the gravity reconstruction is due to the constraint provided by the optical images in the spacecraft-asteroid relative state (Godard et al. 2015). The NavCam will produce these estimates by exploiting image processing algorithms (e.g., Pugliatti et al. 2023c) to track the evolution of the relative state over time. For on-board applications, these optical observables are the asteroid centroids, but other image processing pipelines can be envisaged once images are downlinked to Earth. These techniques, which are still under evaluation, may include feature tracking algorithms, providing the spacecraft relative position and velocity with respect to target landmarks on the surface, silhouette-based techniques, using the illuminated asteroid limb for simultaneous localization and mapping, or other correlation-based methods.

Another contribution that the NavCam can provide is the tracking of boulders and small particles orbiting in proximity to the asteroids. This technique has already been successfully used by the OSIRIS-REx mission, where tracking small boulders ejected from the surface of Bennu (Hergenrother et al. 2020; Liounis et al. 2020) enabled the refinement of the gravity

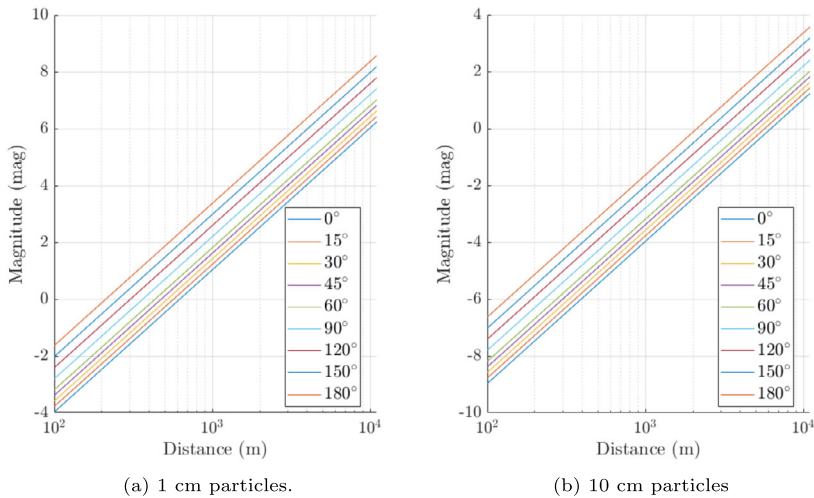


Fig. 20 Apparent magnitude of 1 cm and 10 cm particles as seen from phase angles in the range between 0 and 180 deg

estimation from a 4-degree field to a 9-degree (Scheeres et al. 2019; Chesley et al. 2020). Figure 20 shows the apparent magnitude of 1- and 10-cm particles as a function of distance and phase angle between the particle and the NavCam. This provides quantitative figures to estimate the detectability of particles orbiting in the close proximity of the system, or ejected from the asteroids' surfaces. NavCam's optical measurement of such particles would contribute to improve Didymos's gravity field estimation.

Finally, estimating global properties of Didymos and Dimorphos such as shape and mass will be instrumental to estimating the gravity field of the asteroids. Using analytical functions for polyhedrons and knowing the asteroids' shape and mass (see Sect. 4.1), their constant-density gravity field can be determined together with the corresponding spherical harmonics (SH) coefficients (Werner and Scheeres 1996; Werner 1997). The constant-density SH coefficients can be compared with their OD-estimated counterparts to determine the internal porosity and identify possible density heterogeneity in the asteroid (Scheeres et al. 2020, 2006; Konopliv et al. 2002). While uncertainties in the OD gravity field arise naturally from the estimation process, uncertainties in shape-based estimation arise from both mass and shape uncertainties. These uncertainties should be quantified and compared to assess the fidelity of the shape-based gravity reconstruction (Bercovici et al. 2020; Panicucci et al. 2020). By providing optical images of Didymos and Dimorphos from a variety of geometric and illuminating conditions, the NavCam can potentially reduce the uncertainty in the shape reconstruction with respect to the one obtained from AFC images only, enabling a more robust interpretation of the estimated gravity field.

4.4 Navigation Experiment

The Autonomous Optical Navigation (AutOpNav) Experiment is a technology demonstration aimed at validating the autonomous optical navigation capabilities of Milani in proximity of the Didymos system. Its primary goal is to assess the effectiveness of the image processing (IP) and the navigation algorithms at the core of Milani's on-board Guidance, Navigation, and Control (GNC) system (Pugliatti et al. 2023c). This constitutes one of the

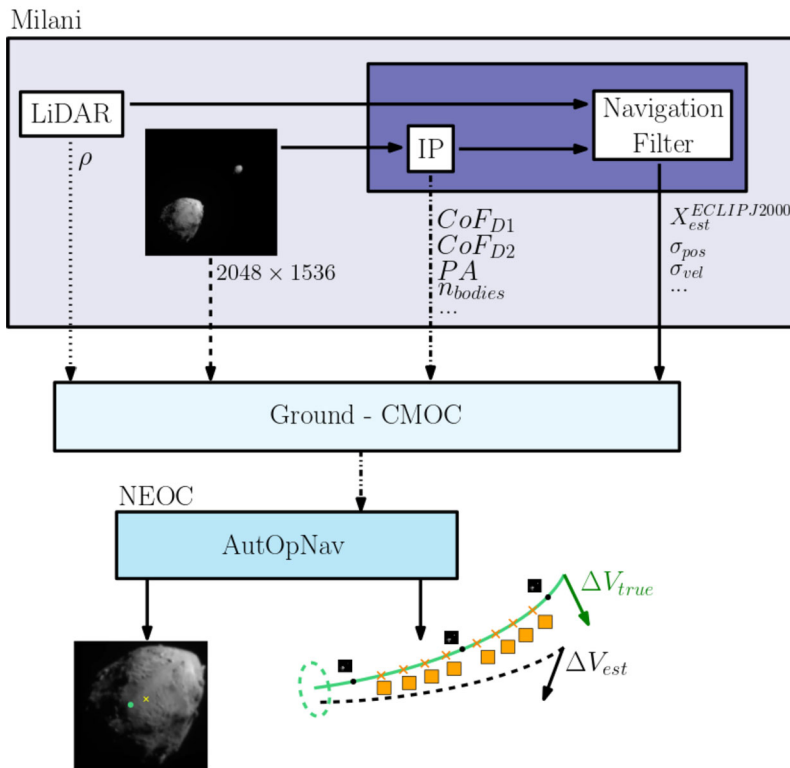


Fig. 21 High-level concept of operations for the AutOpNav experiment

main technological objectives of the mission. Figure 21 shows a schematic overview of the experiment.

A dedicated Navigation Experiment Operation Centre (NEOC) is going to carry out the AutOpNav experiment. The NEOC consists of an on-ground infrastructure designed to collect and exploit the data generated in-flight by Milani's GNC system. The NEOC is located at Politecnico di Milano in Milan (Italy), and is coordinated and operated by the Milani GNC Team at Politecnico di Milano. A schematic architecture of the NEOC is illustrated in Fig. 22. The NEOC will asynchronously gather the GNC data available on ground (including a subset of NavCam images) from the CubeSat Mission Operations Centre (CMOC), which is located at ESA's ESEC centre in Redu (Belgium), and is coordinated and operated by Spacebel. To avoid additional burden on the limited data budget available for the mission, the NEOC has been designed as an opportunistic experiment. Thus, it will rely only on data that is already available on ground, such as those used for the nominal orbit determination (OD) operations.

The dataset available on ground will be augmented with synthetically generated images, in order to reproduce the sequence of pictures actually captured by the spacecraft and used by its on-board algorithms. The geometrical configuration used to generate the synthetic images will be based on the high-accuracy solution produced by the on-ground orbit determination process. The images will be tuned to reproduce the characteristics of the Milani NavCam. Additionally, optical facilities (Panicucci and Toppito 2022; Panicucci et al. 2024) will be employed to capture camera-in-the-loop images to be used in the validation of the

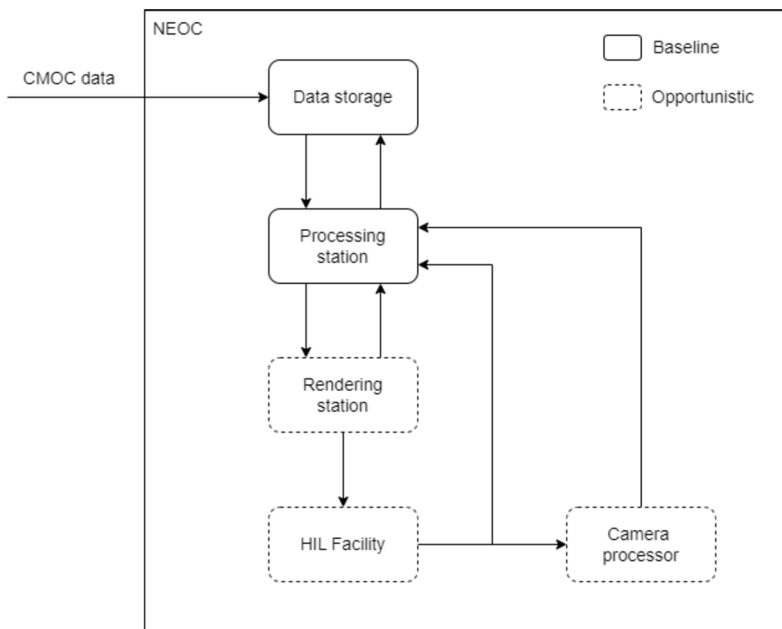
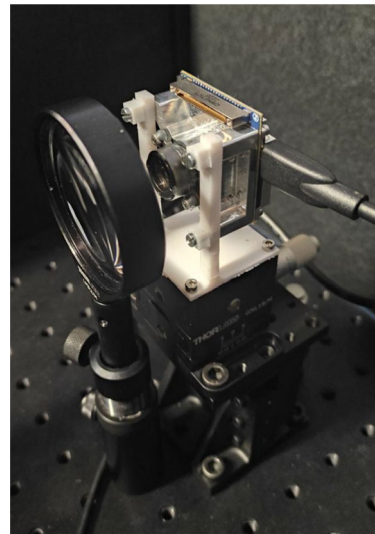


Fig. 22 High-level architecture of the NEOC

Fig. 23 Engineering model of the Milani NavCam mounted in the TinyV3RSE facility



IP and navigation algorithms. Preliminary work to integrate the Milani NavCam with optical facilities and evaluate the IP performance with facility-generated images has already been carried out (Piccolo et al. 2024; Panicucci et al. 2022), also including processor-in-the-loop simulations (Rizza et al. 2022). These tests were used to qualitatively tune IP parameters with facility-generated images acquired with the engineering model of the Milani NavCam. Figure 23 shows the latter mounted in the TinyV3RSE optical testbench (Pan-

Fig. 24 Example of an image of the Didymos system (a) projected in the facility and (b) acquired by the Navcam engineering model



(a)



(b)

icucci et al. 2022), while Fig. 24 compares a synthetic image projected on the facility screen with the corresponding one taken by the Milani NavCam. The synthetic datasets used by the IP and projected in the optical facility have been generated using CORTO (Pugliatti et al. 2023a).

The main activity of the AutOpNav experiment consists in comparing the outcome of the on-board navigation algorithms with a ground-based reference solution to assess their accuracy and performance. Through this comparison, the experiment will validate the performance of the on-board image processing algorithms in terms of errors of the produced measurements. Variations of their implementation and of the tuning parameters could be opportunistically tested on the same dataset for additional insights. This is especially relevant for the data-driven modules employed in the IP (Pugliatti et al. 2024, 2023b) and to inform future developments.

The ground OD will be carried out using landmark-based navigation, which is expected to provide more accurate results with respect to the on-board centroid-based navigation. Therefore, the navigation error will be computed considering the former as the reference solution, and it will be used as the main performance metric. Additionally, the covariance matrix of the on-board estimator can be evaluated to assess its representativeness of the position and velocity uncertainty. Similarly to the IP, different estimators could be tested for their preliminary validation against the baseline performance of the on-board EKF.

The behavior of Milani's GNC state machine will also be evaluated to assess its capability to autonomously switch modes and adapt to varying operational conditions, closely

mirroring the functionality of the state machine on-board the spacecraft while it's flying. Furthermore, there could also be the opportunity to use the results of these assessments to improve the on-board navigation performance during operations. However, the in-flight update of the navigation algorithms is not part of the current baseline of the experiment.

As last stage of the experiment, fictitious trajectory manoeuvres will be computed based on the on-board navigation knowledge and compared to the actual solution produced by the Flight Dynamics operation for the mission. The propagation of the spacecraft trajectory in a simulation environment thus offers a way to evaluate the deviation with respect to the reference trajectory. This process will assess the capability of an autonomous closed-loop guidance cycle as if it was flying on the actual spacecraft.

5 Conclusion

The Milani NavCam offers measurements in the visible spectrum between 400 and 700 nm, and is equipped with a RGB Bayer filter. The camera will provide color images to support several objectives of the Hera/Milani mission, both scientific and technological. Supported scientific investigations include the measure of global properties of Didymos and Dimorphos, the characterization of their surfaces. In this respect, NavCam images will complement and provide context to the observable of the Asteroid Spectral Imager (ASPECT, Näsilä et al. 2025), also carried by the Milani cubesat, which operates between 650 and 2500 nm. In detail, the ASPECT visible channel has a 10-degree FoV and a 1024-pixel square detector, resulting in an iFoV similar to the NavCam, while the spectral bands are strongly complementary, with a small overlap between 650 and 700 nm. The NavCam will be also complementary to the Hera Asteroid Framing Camera (AFC, Vincent et al. 2026) investigation. AFC is a panchromatic camera designed to have a GSD of 50 cm/pixel of both asteroids in Hera Close Operation Phase (Michel et al. 2022), with dedicated flybys of the DART crater reaching as low as 10 cm/pixel. The Milani NavCam can integrate these observations with color images, and with complementary observing geometry. In addition, the NavCam will support gravity field determination, especially during the close fly-by and landing phase. Finally, NavCam images will be used to carry out the autonomous navigation experiment, to support the technological objectives of Milani. In conclusion, the NavCam provides unique observation in terms of both wavelengths and geometry with respect to all other instruments on board Milani and Hera. Its data products support the achievement of Milani and Hera scientific objectives, contributing to the characterization of global properties of asteroids Didymos and Dimorphos, their surface properties as well as the DART impact outcome, and are paramount towards the success of Milani's main technological objective.

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Declarations

Competing Interests The authors declare no competing interests.

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Authors and Affiliations

Fabio Ferrari¹  · **Iosto Fodde**¹ · **Felice Piccolo**¹ · **Carmine Giordano**¹ · **Antonio Rizza**¹ · **Alessia Cremasco**¹ · **Paolo Panicucci**¹ · **Lucia Francesca Civati**¹ · **Pietro Califano**¹ · **Mattia Pugliatti**^{1,2} · **Francesco Topputo**¹ · **Margherita Cardi**³ · **Marco Pavoni**³ · **Daniele Calvi**³ · **Andrea Zanotti**³ · **Filippo Corradino**³ · **Gabor Kovacs**⁴ · **Ernesto Palomba**⁵ · **Fabrizio Dirri**⁵ · **Chiara Gisellu**⁵ · **Marianna Angrisani**⁵ · **Paolo Tortora**^{6,7} · **Riccardo Lasagni Manghi**^{6,7} · **Edoardo Gramigna**^{6,7} · **Michael Küppers**⁸ · **Patrick Michel**^{9,10}

✉ **Fabio Ferrari**
fabio1.ferrari@polimi.it

Iosto Fodde
iosto.fodde@polimi.it

Felice Piccolo
felice.piccolo@polimi.it

Carmine Giordano
carmine.giordano@polimi.it

Antonio Rizza
antonio.rizza@polimi.it

Alessia Cremasco
alessia.cremasco@polimi.it

Paolo Panicucci
paolo.panicucci@polimi.it

Lucia Francesca Civati
luciafrancesca.civati@polimi.it

Pietro Califano
pietro.califano@polimi.it

Mattia Pugliatti
mattia.pugliatti@colorado.edu

Francesco Topputo
francesco.topputo@polimi.it

Margherita Cardi
margherita@tyvak.eu

Marco Pavoni
marco.pavoni@tyvak.eu

Daniele Calvi
daniele.calvi@tyvak.eu

Andrea Zanotti
andrea.zanotti@tyvak.eu

Filippo Corradino
filippo.corradino@tyvak.eu

Gabor Kovacs
kovgab@mogi.bme.hu

Ernesto Palomba
ernesto.palomba@inaf.it

Fabrizio Dirri
fabrizio.dirri@inaf.it

Chiara Gisellu
chiara.gisellu@inaf.it

Marianna Angrisani
marianna.angrisani@inaf.it

Paolo Tortora
paolo.tortora@unibo.it

Riccardo Lasagni Manghi
riccardo.lasagni@unibo.it

Edoardo Gramigna
edoardo.gramigna@unibo.it

Michael Küppers
michael.kueppers@esa.int

Patrick Michel
michelp@oca.eu

¹ Department of Aerospace Science and Technology, Politecnico di Milano, Via La Masa 34, Milan, 20156, Italy

² Colorado Center for Astrodynamics Research, University of Colorado, 3775 Discovery Drive, Boulder, 80303, CO, USA

³ Tyvak International srl, Via Orvieto 19, Torino, 10149, Italy

⁴ Department of Mechatronics, Optics and Mechanical Engineering Informatics, Budapest University of Technology and Economics, Muegyetem rkp. 3, Budapest, H-1111, Hungary

⁵ INAF-IAPS, Via del Fosso del Cavaliere 100, Rome, 00133, Italy

⁶ Dipartimento di Ingegneria Industriale, Alma Mater Studiorum Università di Bologna, via Fontanelle 40, Forlì, 47121, Italy

⁷ Centro Interdipartimentale di Ricerca Industriale Aerospaziale, Alma Mater Studiorum Università di Bologna, via Baldassarre Carnaccini 12, Forlì, 47121, Italy

⁸ ESAC, European Space Agency, Camino Bajo del Castillo, Villafranca del Castillo, Madrid, Spain

⁹ Université Côte d'Azur, Observatoire de la Côte d'Azur, CNRS, Laboratoire Lagrange, Nice, France

¹⁰ The University of Tokyo, Department of Systems Innovation, School of Engineering, Tokyo, Japan